STACKS, DISCRIMINANTS: LEMMA-TRACE-BASE-CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 893-908 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 910-952, the complete author proof (41 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_discriminant.json (pb=2102, stmt 893-908, proof 910-952).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BT5: sources/stacks-discriminant/pinned-original-source.tex, source lines 493–497. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT2: sources/stacks-discriminant/pinned-original-source.tex, source lines 380–387. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a flat quasi-finite map of Noetherian rings. Let $\tau \in \omega_{B/A}$ be a trace element.*

1. *If $A \to A_1$ is a map with A_1 Noetherian, then with $B_1 = A_1 \otimes_A B$ the image of τ in ω_{B_1/A_1} is a trace element.*
2. *If $A = R_f$ for some ring R and $f \in R$, then τ is a trace element in $\omega_{B/R}$.*
3. *If $g \in B$, then the image of τ in $\omega_{B_g/A}$ is a trace element.*
4. *If $B = B_1 \times B_2$, then τ maps to a trace element in both $\omega_{B_1/A}$ and $\omega_{B_2/A}$.*

Proof. Part (1) is a formal consequence of the definition.

Statement (2) makes sense because $\omega_{B/R} = \omega_{B/A}$ by Lemma 0BT2. Denote τ' the element τ but viewed as an element of $\omega_{B/R}$. To see that (2) is true suppose that we have $R \to R_1$ with R_1 Noetherian and a product decomposition $B \otimes_R R_1 = C \times D$ with $R_1 \to C$ finite. Then with $A_1 = (R_1)_f$ we see that $B \otimes_A A_1 = C \times D$. Since $R_1 \to C$ is finite, a fortiori $A_1 \to C$ is finite. Hence we can use the defining property of τ to get the corresponding property of τ'.

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Statement (3) makes sense because $\omega_{B_g/A} = (\omega_{B/A})_g$ by Lemma 0BT2. The proof is similar to the proof of (2). Suppose we have $A \to A_1$ with A_1 Noetherian and a product decomposition $B_g \otimes_A A_1 = C \times D$ with $A_1 \to C$ finite. Set $B_1 = B \otimes_A A_1$. Then $\mathrm{Spec}(C) \to \mathrm{Spec}(B_1)$ is an open immersion as $B_g \otimes_A A_1 = (B_1)_g$ and the image is closed because $B_1 \to C$ is finite (as $A_1 \to C$ is finite). Thus we see that $B_1 = C \times D_1$ and $D = (D_1)_g$. Then we can use the defining property of τ to get the corresponding property for the image of τ in $\omega_{B_g/A}$.

Statement (4) makes sense because $\omega_{B/A} = \omega_{B_1/A} \times \omega_{B_2/A}$ by Lemma 0BT5. Suppose we have $A \to A'$ with A' Noetherian and a product decomposition $B \otimes_A A' = C \times D$ with $A' \to C$ finite. Then it is clear that we can refine this product decomposition into $B \otimes_A A' = C_1 \times C_2 \times D_1 \times D_2$ with $A' \to C_i$ finite such that $B_i \otimes_A A' = C_i \times D_i$. Then we can use the defining property of τ to get the corresponding property for the image of τ in $\omega_{B_i/A}$. This uses the obvious fact that $\mathrm{Trace}_{C/A'} = (\mathrm{Trace}_{C_1/A'}, \mathrm{Trace}_{C_2/A'})$ under the decomposition $\omega_{C/A'} = \omega_{C_1/A'} \times \omega_{C_2/A'}$. $\square$